

# Ghost Gutzwiller for the Half-Filled Hubbard Dimer by Direct Minimization (No Impurity Problem): Equations and Implementation of `not_imp_gga_dimer.py`

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## Contents

|           |                                                                                                         |          |
|-----------|---------------------------------------------------------------------------------------------------------|----------|
| <b>1</b>  | <b>The problem</b>                                                                                      | <b>1</b> |
| <b>2</b>  | <b>The variational functional</b>                                                                       | <b>2</b> |
| <b>3</b>  | <b>Variables and parametrization</b>                                                                    | <b>3</b> |
| <b>4</b>  | <b>Objective, constraints and analytic gradients</b>                                                    | <b>4</b> |
| <b>5</b>  | <b>Minimization</b>                                                                                     | <b>5</b> |
| <b>6</b>  | <b>Recovery of the multipliers and consistency checks</b>                                               | <b>5</b> |
| <b>7</b>  | <b>Second-order test</b>                                                                                | <b>6</b> |
| <b>8</b>  | <b>Results (<math>U/t = 2</math>, <math>N_g = 2</math>) and status</b>                                  | <b>6</b> |
| <b>9</b>  | <b>Usage and correspondence with <code>gga_dimer.py</code></b>                                          | <b>8</b> |
| <b>10</b> | <b>Reconstruction of <math>R</math> and <math>\lambda</math> from <math>x</math> and <math>X</math></b> | <b>8</b> |

**Purpose of this note.** `gga_dimer.py` solves the ghost Gutzwiller (gGA) equations of the Hubbard dimer by the iterative embedding cycle of `code.tex`: an impurity Hamiltonian  $H_{\text{imp}}(V, \lambda^c)$  is diagonalized at every iteration. `not_imp_gga_dimer.py` is an *alternative* solver for *the same problem*: no impurity Hamiltonian, no  $V^I$ , no  $\lambda^{I,c}$  and no fixed-point loop appear. Following Tagliente, Pasqua and Fabrizio (arXiv:2607.27156, `NotImpurity.pdf`, Eqs. 7 and 9–12), the variational energy functional is minimized *directly* over the impurity wavefunctions  $|\phi^I\rangle$  and the quasiparticle Slater determinant, subject to the density constraints. The notation is that of `code.tex` ( $N_{\text{eff}} = 1 + N_g$ ,  $R^I$ ,  $\Delta^{II}$ ,  $D^I$ ,  $\Delta^{I,\text{bb}}$ ,  $\rho$ ,  $\lambda^I$ ,  $S(\Delta)$ ,  $\mathbf{I}$ ,  $X_{ab}$ , ...); equations of `code.tex` are cited by name, not by number.

## 1 The problem

**Model.** Exactly as in `code.tex`: the half-filled two-site Hubbard dimer with hopping  $-t$  and local Hamiltonian  $H_{\text{loc}} = U n_{\uparrow} n_{\downarrow} - \mu (n_{\uparrow} + n_{\downarrow})$ ,  $\mu = U/2$  (it differs from the  $\frac{U}{2}(n-1)^2$  of the

reference paper by the constant  $U/2$  per site, which is kept so that energies are directly comparable with `gga_dimer.py`).  $N_{\text{phys}} = 1$  physical orbital per fragment,  $N_{\text{eff}} = 1 + N_g$  quasiparticle (bath) orbitals per fragment and spin;  $N_g$  is even, so  $N_{\text{eff}}$  is odd.

**Symmetry class.** As in `gga_dimer.py`: paramagnetic (spin-restricted), the two fragments  $I = 0, 1$  *not* assumed equal.

- The impurity wavefunction  $|\phi^I\rangle$  is a *real* vector in the half-filled, spin-balanced sector  $N_{\uparrow} = N_{\downarrow} = (1 + N_{\text{eff}})/2$  of the embedding Fock space (dimension  $4^{1+N_{\text{eff}}}$ ), a sector of dimension

$$m = \binom{1 + N_{\text{eff}}}{(1 + N_{\text{eff}})/2}^2 \quad (m = 36 \text{ for } N_g = 2). \quad (1)$$

All operators entering below conserve  $N_{\uparrow}$  and  $N_{\downarrow}$  separately, so their restriction to this sector is exact (no leakage).

- The quasiparticle Slater determinant  $|\psi_*\rangle$  has one 1-RDM  $\rho \in \mathbb{R}^{2N_{\text{eff}} \times 2N_{\text{eff}}}$  (per spin, the same for both), a rank- $N_{\text{eff}}$  projector: half filling, i.e.  $N_{\text{eff}}$  of the  $2N_{\text{eff}}$  levels are occupied (the filling rule of `code.tex`).

Blocks of  $\rho$  are written  $\rho^{IJ}$  ( $\rho^{II}$  is  $\Delta_{\text{qp}}^{II}$  of `code.tex`).

## 2 The variational functional

**Energy (Eq. 11 of the reference).** Specialized to the dimer and summed over spin and over the two bonds,

$$E(\rho, \phi^0, \phi^1) = -4t (R^0)^T \rho^{01} R^1 + \sum_{I=0,1} \langle \phi^I | H_{\text{loc}} | \phi^I \rangle. \quad (2)$$

The factor 4 is  $2$  (spin)  $\times 2$  (the two ordered fragment pairs), the same factor as in the variational energy of `code.tex`. Here  $R^I$  is *not* an independent variable: it is the function  $R(\phi^I)$  of Eqs. 9–10 of the reference,

$$R^I = S(\Delta^I)^{-1} D^I, \quad \Delta^I \equiv \mathbf{I} - \Delta^{I,\text{bb}}, \quad S(\Delta) = \sqrt{\Delta(\mathbf{I} - \Delta) + \varepsilon \mathbf{I}}, \quad (3)$$

with the two expectation values over the impurity state (spin averages)

$$\Delta_{ab}^{I,\text{bb}} = \frac{1}{2} (\langle d_{a\uparrow}^\dagger d_{b\uparrow} \rangle + \langle d_{a\downarrow}^\dagger d_{b\downarrow} \rangle)_{\phi^I}, \quad D_a^I = \frac{1}{2} (\langle d_{a\uparrow}^\dagger c_\uparrow \rangle + \langle d_{a\downarrow}^\dagger c_\downarrow \rangle)_{\phi^I}. \quad (4)$$

These are the same  $\Delta^{I,\text{bb}}$  and  $D^I$  as in the impurity section of `code.tex`;  $S(\Delta)$  is the shifted square root of `code.tex` ( $\varepsilon = 10^{-9}$  by default, added to the eigenvalues of  $\Delta(\mathbf{I} - \Delta)$ ). Since  $\Delta \mapsto \mathbf{I} - \Delta$  leaves  $\Delta(\mathbf{I} - \Delta)$  unchanged,  $S(\mathbf{I} - \Delta^{I,\text{bb}}) = S(\Delta^{I,\text{bb}})$ , i.e. Eq. (3) is the reference's  $R = Q S^{-1}$  with  $S^2 = \Delta_\phi(\mathbf{I} - \Delta_\phi)$ ,  $\Delta_\phi = \Delta^{I,\text{bb}}$ ,  $Q = D$ .

**Constraints (Eq. 7 of the reference).**

$$\langle \phi^I | \phi^I \rangle = 1, \quad \rho^{II} = \mathbf{I} - \Delta^{I,\text{bb}}(\phi^I) \quad (I = 0, 1), \quad (5)$$

the second being the requirement that the local density matrix of the Slater determinant equal the particle–hole transform of the bath density matrix of the impurity state.

**The functional  $F$  (Eq. 12) and the Lagrangian actually used.** Eq. 12 of the reference is  $F = E - E_* \langle \psi_* | \psi_* \rangle - E_\phi \langle \phi | \phi \rangle + \text{Tr}\{\hat{\lambda}[\mathbf{I} - \hat{\Delta}_\phi - \hat{\Delta}_*]\}$ : energy, two normalizations, and a single matrix multiplier  $\hat{\lambda}$  for the density constraint. With our spin factors and eliminating the two normalizations (see below), the saddle point of  $F$  is the stationary point of

$$\mathcal{L} = E(\rho, \phi^0, \phi^1) + \sum_I 2 \text{Tr} \left[ \lambda^I (\mathbf{I} - \Delta^{I, \text{bb}}(\phi^I) - \rho^I) \right], \quad (6)$$

where  $\lambda^I$  is the same  $\lambda^I$  as in the quasiparticle Hamiltonian  $H_{\text{qp}} = \begin{bmatrix} -\lambda^0 & -tR^0R^{1T} \\ -tR^1R^{0T} & -\lambda^1 \end{bmatrix}$  of `code.tex`. Compared with the Lagrangian of `code.tex` (which also carries  $V^I$ ,  $\lambda^{I,c}$  and the independent variables  $R^I, \Delta^I$ ), the constraints associated with  $V^I$  ( $D^I = S(\Delta^I)R^I$ ) and with  $\lambda^{I,c}$  ( $\Delta^I + \Delta^{I, \text{bb}} = \mathbf{I}$ ) have been *substituted*:  $R^I = R(\phi^I)$  and  $\Delta^I = \mathbf{I} - \Delta^{I, \text{bb}}$ . This is the simplification of the direct approach.

### 3 Variables and parametrization

**Unknowns.** The unconstrained (redundant) variables are

$$z = (x^0, x^1, X), \quad x^I \in \mathbb{R}^m, \quad X \in \mathbb{R}^{2N_{\text{eff}} \times N_{\text{eff}}}, \quad (7)$$

with the two parametrizations

$$\phi^I = \frac{x^I}{\|x^I\|}, \quad \rho = X(X^T X)^{-1} X^T. \quad (8)$$

The first automatically enforces  $\langle \phi^I | \phi^I \rangle = 1$  (so the norm multipliers  $E_\phi$  never appear); the second is the orthogonal projector on the column space of  $X$ , a rank- $N_{\text{eff}}$  projector for any full-rank  $X$  (so  $|\psi_*\rangle$  is automatically a half-filled Slater determinant and  $E_*$  never appears). The redundancy is a gauge:  $x^I \rightarrow c x^I$  and  $X \rightarrow XA$  ( $A \in GL(N_{\text{eff}})$ ) leave everything invariant. The number of variables is  $n_z = 2m + 2N_{\text{eff}}^2$  ( $= 90$  for  $N_g = 2$ ).

**Operator matrices (precomputed once).** From the sparse Jordan–Wigner operators of `gga_dimer.py` (build `impurity_ops`, mode order  $c_\uparrow, c_\downarrow, d_{0\uparrow}, d_{0\downarrow}, \dots$ ) the following  $m \times m$  real matrices are built by restriction to the sector:

$$\begin{aligned} H_{\text{loc}}^s &= \text{sector}(H_{\text{loc}}), \\ \mathcal{Y}_a &= \text{sector}(d_{a\uparrow}^\dagger c_\uparrow + c_\uparrow^\dagger d_{a\uparrow} + d_{a\downarrow}^\dagger c_\downarrow + c_\downarrow^\dagger d_{a\downarrow}), \\ B_{ab} &= \frac{1}{2} \text{sector}(d_{a\uparrow}^\dagger d_{b\uparrow} + d_{a\downarrow}^\dagger d_{b\downarrow}) \quad \text{symmetrized in } (a, b), \end{aligned} \quad (9)$$

together with  $n_\uparrow n_\downarrow$  and  $n_\uparrow + n_\downarrow$  for the observables. For a real  $\phi$  (so  $\langle d^\dagger c \rangle = \langle c^\dagger d \rangle$ ) the expectation values of Eqs. (4) become plain quadratic forms:

$$\Delta_{ab}^{I, \text{bb}} = \phi^T B_{ab} \phi, \quad D_a^I = \frac{1}{4} \phi^T \mathcal{Y}_a \phi, \quad E_{\text{loc}}^I = \phi^T H_{\text{loc}}^s \phi. \quad (10)$$

(The  $\frac{1}{4}$  is  $\frac{1}{2}$  from the spin average times  $\frac{1}{2}$  from  $\mathcal{Y}_a$  containing both  $d^\dagger c$  and its transpose, for each spin.) This is the same evaluation as in `impurity_solve` of `gga_dimer.py`, except that the state  $\phi$  is now a *variable* instead of the ground state of  $H_{\text{imp}}$ .

## 4 Objective, constraints and analytic gradients

Everything is differentiated analytically (checked against central finite differences by `--check-gradients`: agreement  $\sim 2 \times 10^{-10}$  for both  $E$  and the constraints).

**Evaluation of  $E$  (energy).** For each fragment:  $\phi^I, \Delta^{I,\text{bb}}, D^I$  from Eqs. (8), (10); diagonalize  $\Delta^I = \mathbf{I} - \Delta^{I,\text{bb}} = W \text{diag}(d_i) W^T$ ;  $m_i = d_i(1 - d_i) + \varepsilon$ ,  $f_i = m_i^{-1/2}$ ;  $R^I = W[f \circ (W^T D^I)]$ ; then Eq. (2). (A valid 1-RDM has all  $d_i \in [0, 1]$ , so up to rounding  $m_i > 0$ ; the shift  $\varepsilon$  covers rounding of  $d_i$  to slightly negative values.)

**Gradient with respect to  $\rho$  and  $X$ .**

$$dE = \text{Tr}[G_\rho d\rho] + \dots, \quad G_\rho = \begin{pmatrix} 0 & -2t R^0 R^{1T} \\ -2t R^1 R^{0T} & 0 \end{pmatrix}, \quad (11)$$

and, from  $d\rho = P_\perp dX Y X^T + X Y dX^T P_\perp$  with  $Y = (X^T X)^{-1}$ ,  $P_\perp = \mathbf{I} - \rho$ ,

$$\frac{\partial E}{\partial X} = 2 P_\perp G_\rho X Y. \quad (12)$$

**Gradient with respect to  $R^I$  and through  $R(\phi^I)$ .** The partial derivatives with respect to the (dependent) vectors  $R^I$  are

$$g_{R^0} = -4t \rho^{01} R^1, \quad g_{R^1} = -4t (\rho^{01})^T R^0. \quad (13)$$

Write  $R = f(M)D$  with  $M = \Delta(\mathbf{I} - \Delta) + \varepsilon \mathbf{I}$  and  $f(m) = m^{-1/2}$ . In the eigenbasis of  $\Delta$ ,  $\tilde{u} = W^T g_R$ ,  $\tilde{v} = W^T D$ , the Daletskii–Krein derivative of  $f(M)$  is the divided-difference matrix

$$\mathcal{F}_{ij} = \frac{f(m_i) - f(m_j)}{m_i - m_j} = -\frac{1}{s_i s_j (s_i + s_j)}, \quad s_i = \sqrt{m_i}, \quad (14)$$

which is written in the cancellation-free form on the right (valid also for  $m_i = m_j$ ). Since  $dM_{ij} = (1 - d_i - d_j) d\Delta_{ij}$  in this basis and  $d\Delta = -d\Delta^{I,\text{bb}}$ ,

$$g_D = f(M) g_R = W (f \circ \tilde{u}), \quad \Gamma = \text{sym}[(\tilde{u} \tilde{v}^T) \circ \mathcal{F}], \quad g_{\text{bb}} = -W [\Gamma \circ (1 - d_i - d_j)] W^T. \quad (15)$$

(This is the same structure as the function `gradS` of `code.tex`, here applied as an adjoint.) Collecting the three routes by which  $\phi^I$  enters (through  $\Delta^{I,\text{bb}}$ , through  $D^I$ , and through  $E_{\text{loc}}$ ),

$$g_\phi = 2 \sum_{ab} (g_{\text{bb}})_{ab} B_{ab} \phi + \frac{1}{2} \sum_a (g_D)_a \mathcal{Y}_a \phi + 2 H_{\text{loc}}^s \phi, \quad \frac{\partial E}{\partial x^I} = \frac{1}{\|x^I\|} (g_\phi - (\phi \cdot g_\phi) \phi), \quad (16)$$

the last step being the projection through  $\phi = x/\|x\|$ .

**Constraints and Jacobian (constraints, constraint\_jac).** The independent entries ( $a \leq b$ ) of the two symmetric matrix equations of Eq. (5),

$$c_{ab}^I(z) = (\rho^{II} + \Delta^{I,\text{bb}} - \mathbf{I})_{ab} = 0, \quad a \leq b, \quad I = 0, 1 \quad (17)$$

give  $n_c = N_{\text{eff}}(N_{\text{eff}} + 1)$  equations ( $= 12$  for  $N_g = 2$ ). Their gradients are

$$\frac{\partial c_{ab}^I}{\partial x^I} = \frac{1}{\|x^I\|} (\mathbf{1} - \phi \phi^T) 2B_{ab} \phi, \quad \frac{\partial c_{ab}^I}{\partial X} = 2 P_{\perp} G^{(I,ab)} X Y, \quad (18)$$

where  $G^{(I,ab)}$  is the symmetric matrix that has  $\frac{1}{2}$  at positions  $(a, b), (b, a)$  (or 1 at  $(a, a)$  if  $a = b$ ) inside the  $I$ -th diagonal block and zero elsewhere. (For  $N_g = 2$  the constraint Jacobian has full rank 12 at the solutions examined, backward and forward.)

## 5 Minimization

**Algorithm.** The constrained problem

$$\min_z E(z) \quad \text{subject to} \quad c(z) = 0 \quad (19)$$

is solved by sequential quadratic programming (`scipy.optimize.minimize`, `method="SLSQP"`, `ftol` =  $10^{-13}$ , up to 1000 iterations), passing the analytic gradient of  $E$  and the analytic Jacobian of  $c$ . A run is accepted as *feasible* if  $\max |c| < 10^{-8}$ ; among several starts the lowest-energy feasible result is kept. No impurity Hamiltonian is ever diagonalized in the minimization.

**Starting points.**

1. *Random start* (`random_start`):  $x^I$  with independent Gaussian components;  $X$  = the occupied orbitals of  $H_{\text{qp}}$  built from the default cold-start  $R^0 = R^1 = (0.9, 0.05, 0.065, \dots)$ ,  $\lambda = 0$  of `gga_dimer.py`, plus  $0.05 \times$  Gaussian noise.
2. *From an  $(R, \lambda)$  guess* (`start_from_R_lambda`; files `R.in`, `L.in` in the format of `gga_dimer.py`):  $X$  = the  $N_{\text{eff}}$  lowest eigenvectors of  $H_{\text{qp}}(R, \lambda)$ ;  $\Delta^{II} = \rho^{II}$ ,  $V^I$  from the Eq. 7 of `code.tex` and  $\lambda^{I,c}$  from its Eq. 8; then the lowest 16 eigenstates of the sector matrix  $H_{\text{loc}}^s + \sum_a V_a^I \mathcal{Y}_a + \sum_{ab} \lambda_{ab}^{I,c} 2B_{ab}$  are examined and the one whose bath density best satisfies  $\Delta^{I,\text{bb}} = \mathbf{I} - \Delta^{II}$  is taken as  $x^I$ . (This single diagonalization is used *only* to construct the starting  $\phi^I$ ; choosing among the low-lying eigenstates matters for the “forward” seed, whose impurity ground state has swapped ghost occupations and would otherwise violate the constraints by  $\approx 0.97$ , after which SLSQP restores feasibility far away, at  $E \approx -0.80$ .)

## 6 Recovery of the multipliers and consistency checks

At a solution the multipliers of Eq. (6) are recovered, and the point is checked against the equations of the embedding approach of `code.tex` (`analyse`).

**Stationarity conditions.**  $\partial \mathcal{L} / \partial \rho = 0$  on the manifold of projectors and  $\partial \mathcal{L} / \partial \phi^I = 0$  on the unit sphere give, with  $K = \begin{bmatrix} 0 & -tR^0 R^{1T} \\ -tR^1 R^{0T} & 0 \end{bmatrix}$  the hopping part of  $H_{\text{qp}}$  and  $\Lambda = \text{diag}(\lambda^0, \lambda^1)$ ,

$$[K - \Lambda, \rho] = 0 \quad (\text{i.e. } H_{\text{qp}} = K - \Lambda, \rho \text{ an eigenprojector}), \quad (20)$$

$$\nabla_{\phi} E - 4 \sum_{ab} \lambda_{ab}^I B_{ab} \phi^I = \mu_I \phi^I \quad (I = 0, 1). \quad (21)$$

Both are *linear* in the  $2 N_{\text{eff}}(N_{\text{eff}} + 1)/2$  independent entries of  $\lambda^0, \lambda^1$ . They are solved *jointly* by linear least squares. (Using Eq. (21) alone is not sufficient: when a ghost is inert the vectors

$B_{ab}\phi$  are linearly dependent, its minimum-norm solution is not unique and need not make  $\rho$  an eigenprojector of  $H_{\text{qp}}$ ; with the joint system the rank is 12/12 for  $N_g = 2$ .) The residuals of the joint system are reported as “KKT:  $\phi$  stationarity” and “KKT:  $[H_{\text{qp}}, \rho]$ ”.

**Quasiparticle energies.** With the recovered  $\lambda^I$ ,  $H_{\text{qp}}(R, \lambda)$  is built and diagonalized; its  $2N_{\text{eff}}$  eigenvalues are the quasiparticle energies `eqp` of the scans of `code.tex`.

**Check against the embedding equations.** With  $R^I = R(\phi^I)$  and the recovered  $\lambda^I$ ,  $V^I$  and  $\lambda^{I,c}$  are computed from Eqs. 7 and 8 of `code.tex`, and the residual

$$\|(H_{\text{imp}}^s - \langle \phi^I | H_{\text{imp}}^s | \phi^I \rangle) \phi^I\|_{\infty}, \quad H_{\text{imp}}^s = H_{\text{loc}}^s + \sum_a V_a^I \mathcal{Y}_a + \sum_{ab} \lambda_{ab}^{I,c} 2B_{ab}, \quad (22)$$

is reported (“ $H_{\text{imp}}\phi = E\phi$ ”). A small value shows that  $\phi^I$  is an eigenstate of the embedding Hamiltonian of `code.tex` built from the solution, i.e. that the solution of the direct minimization also solves the embedding equations. The observables  $Z^I = R^{IT}R^I$ , double occupancy and  $n_{\text{phys}}$  are evaluated from Eqs. (10).

## 7 Second-order test

For a stationary point with full-rank constraints, a strict local minimum of the constrained energy requires the Hessian of the Lagrangian to be positive definite on the tangent space of the constraints. `hessian_analysis` computes it as follows.

1. Multipliers  $\mu$  of the constraints from  $\nabla E + J^T \mu = 0$  (least squares),  $J = \partial c / \partial z$ ; residual reported.
2. Hessian  $H$  of  $E + \mu \cdot c$  with respect to all  $n_z$  variables by central differences (step  $h = 10^{-6}$ ) of the analytic gradient  $\nabla E + J(z)^T \mu$ , symmetrized.
3. Tangent space = null space of  $J$  (dimension  $n_z - n_c = 78$  for  $N_g = 2$ ); eigenvalues of the reduced Hessian  $Z_t^T H Z_t$ .

The redundancies of the parametrization give exact zero modes: 2 (norms of  $x^I$ ) +  $N_{\text{eff}}^2$  ( $X \rightarrow XA$ ) +  $N_{\text{eff}}(N_{\text{eff}} - 1)$  (rotations of the ghost basis of the two fragments), i.e. 17 for  $N_g = 2$ . A strict minimum has all other 61 eigenvalues positive.

## 8 Results ( $U/t = 2$ , $N_g = 2$ ) and status

The two  $U/t = 2$  solutions discussed in `code.tex` (“backward” and “forward”, seeds in `examples/`) are used as tests.

|                                                                  | backward                                                 | forward                                                  |
|------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| $E_{\text{var}}/t$ (this code)                                   | -3.1256614043                                            | -3.1249999925                                            |
| $E_{\text{var}}/t$ ( <code>gga_dimer.py</code> )                 | -3.1256614043                                            | -3.12500000                                              |
| $Z^0 = Z^1$                                                      | 0.99991                                                  | 0.9375                                                   |
| $\max  c $                                                       | $6 \times 10^{-15}$                                      | $8 \times 10^{-15}$                                      |
| KKT: $\phi / [H_{\text{qp}}, \rho] / H_{\text{imp}}\phi = E\phi$ | $3 \times 10^{-7} / 8 \times 10^{-9} / 1 \times 10^{-7}$ | $6 \times 10^{-7} / 6 \times 10^{-8} / 3 \times 10^{-7}$ |
| Hessian: negative / zero / positive                              | 0 / 17 / 61                                              | 0 / 20 / 58                                              |
| smallest positive eigenvalue                                     | 2.90                                                     | 0.0060                                                   |

Table 1: Solutions of the direct minimization at  $U/t = 2$ ,  $N_g = 2$ . Backward: started from the  $(R, \lambda)$  seed of `examples/U2.backward`. Forward: converged from a random start (the seed of `examples/U2.forward` leads to the backward solution, see below). The energies agree with `gga_dimer.py`.

- **Backward** is reproduced to  $10^{-12}$  and is a strict local minimum (the 61 positive eigenvalues are all the non-gauge ones; the softest is 2.90).
- **Forward:** unlike the fixed point of `gga_dimer.py` (where  $\rho^{II}$  and  $\mathbf{I} - \Delta^{I,\text{bb}}$  differ by  $\approx 0.97$  with swapped ghost occupations), here  $\rho^{II}$  and  $\mathbf{I} - \Delta^{I,\text{bb}}$  coincide entry by entry ( $10^{-12}$ ), both with eigenvalues  $\{0, \frac{1}{2}, 1\}$ : the constraints of Eq. (5) hold. Multipliers exist (KKT residuals  $\lesssim 10^{-6}$ ). The reduced Hessian has no negative eigenvalue but 3 zero modes in addition to the 17 gauge ones, and its positive eigenvalues are small (0.0060 smallest). Along all 20 zero-mode directions, followed on feasible curves,  $E$  never decreases and grows as  $s^4$  (about  $3 \times 10^{-6}$  at  $s = 0.2$ ); thus the second-order test is *inconclusive along those 3 directions*. The origin of these extra flat directions was not investigated. An independent empirical test (random displacements projected back on the constraints, 4000 of them, steps  $10^{-3}$ –0.1) finds only energy increases growing as  $s^2$  with coefficient  $\approx 7 \times 10^{-3}$ .
- **The forward seed does not stay on the forward minimum.** A single SLSQP pass started from the forward seed stops early ( $E = -3.12499887$ , projected gradient  $4 \times 10^{-3}$ , descent directions with  $\Delta E \simeq -8 \times 10^{-5}$  exist). A second pass continues downhill, *not* to the forward minimum  $-3.1249999925$  but to the lower ghost-assisted (backward) solution  $-3.1256614043$ , and further passes stay there. The forward minimum listed in Table 1 was reached from a random start; it is a local minimum with a small basin, and the seed lies outside it. Consequently the “forward” U-scan of `scan_not_imp_U.py`, seeded from that point, converges to the ghost-assisted branch at  $U/t = 2$  (it stays on the plain-Gutzwiller branch only for  $U/t \lesssim 0.8$ ).

**Limitation: random starts.** From random starts the minimization always reaches the *higher* boundary minima with inert ghosts, never the ghost-assisted solutions: 120/120 starts give  $E = -3.125$  at  $U/t = 2$  (versus  $-3.125661$ ), 30/30 give  $-10.0$  at  $U/t = 10$  (versus  $-10.101314$  from `gga_dimer.py`), 30/30 give  $-6.0$  at  $U/t = 6$ ; at  $U/t = 4$  the minima are  $-4.5$  and  $-4.0$ . Perturbing the boundary minimum by up to 0.3 in the variables and re-minimizing also returns to it. The ghost-assisted solutions are reached only from a seed close to them (as with the backward seed above). Making the direct minimization find them without a seed (for example by continuation in  $U$ , warm-starting  $(x^I, X)$  from the previous  $U$ ) has *not* been done.

## 9 Usage and correspondence with `gga_dimer.py`

```
python3 not_imp_gga_dimer.py --U <U/t> --Ng <even> [--t 1.0] [--nstarts 10] [--seed 0]
    [--R-file R.in --lam-file L.in] # seeded start (else random)
    [--max-iter 1000] [--shift 1e-9] [--quiet]
    [--omega <w_max> --eta <eta> --domega <step>] # spectral
    function
python3 not_imp_gga_dimer.py --check-gradients [--Ng 2 --U 2] # analytic vs finite diff.
```

It prints  $Z^I$ ,  $E_{\text{var}}/t$ , the double occupancy, the constraint violation, the three KKT residuals and `eqp`, and writes `not_imp_Ng*.U*_{R,lambda,lambda_c}.dat` in the layout of `gga_dimer.py`'s `save_results`.

**Spectral function.** If `--omega`, `--eta` and `--domega` are all given (any subset is an error), the spectral function of the solution is computed exactly as in the post-processing section of `code.tex`:  $G(\omega) = [(\omega + i\eta)\mathbf{I} - H_{\text{qp}}]^{-1}$  with the  $H_{\text{qp}}$  of the reconstructed  $(R, \lambda)$  (Section 10), then  $G_{\text{phys}} = R_{\text{mat}}^T G R_{\text{mat}}$  with  $R_{\text{mat}} = \text{diag}(R^0, R^1)$  and  $A(\omega) = -\frac{2}{\pi} \text{Im Tr } G_{\text{phys}}(\omega)$  on the grid  $-\omega_{\text{max}}, \dots, \omega_{\text{max}}$  in steps of  $d\omega$ , written to `A.omega.txt`; the routine is the one of `gga_dimer.py` (`compute_spectral_function`), and the sum rule  $\int A d\omega \rightarrow 4$  is printed. The only input from the minimization is  $(R, \lambda)$ , so  $A(\omega)$  inherits the ambiguity of the nearly free direction of  $\lambda$  described in Section 10; at  $U/t = 10$  (the `examples/U10` seed,  $\omega_{\text{max}} = 15$ ,  $\eta = d\omega = 0.05$ ) it gives  $\int A d\omega = 3.9615$  and agrees with the  $A(\omega)$  of `gga_dimer.py` for the same case to  $6 \times 10^{-5}$  (maximum of  $A$ : 6.48).

|                                                            | <code>gga_dimer.py</code> (embedding)                            | <code>not_imp_gga_dimer.py</code> (direct)   |
|------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------|
| unknowns                                                   | $R^I, \lambda^I$ (and $V^I, \lambda^{I,c}, \Delta^{II}$ derived) | $x^I$ ( $\phi^I$ ), $X$ ( $\rho$ )           |
| $R^I$                                                      | output of the impurity solve,<br>$S(\Delta^{\text{new}})^{-1}D$  | function of $\phi^I$ , Eq. (3)               |
| $V^I, \lambda^{I,c}$                                       | Eqs. 7, 8 of <code>code.tex</code> , needed in every iteration   | not used; only in the a-posteriori check     |
| impurity ground state                                      | diagonalized every iteration                                     | never (only for a seeded start)              |
| constraint $\rho^{II} = \mathbf{I} - \Delta^{I,\text{bb}}$ | <i>not enforced</i> by the convergence test                      | enforced ( $\max  c  \sim 10^{-14}$ )        |
| solver                                                     | damped fixed-point iteration                                     | SLSQP on $E$ with equality constraints       |
| result type                                                | fixed point of the iteration                                     | constrained    local    minimum<br>(checked) |

Table 2: Correspondence between the two solvers.

## 10 Reconstruction of $R$ and $\lambda$ from $x$ and $X$

The unknowns of the direct minimization are only  $z = (x^0, x^1, X)$ ; neither  $R^I$  nor  $\lambda^I$  is a variable. Both are reconstructed from a solution in the function `analyse`, in the following order. Let  $n_s = N_{\text{eff}}(N_{\text{eff}} + 1)/2$  (the number of independent entries of a symmetric  $N_{\text{eff}} \times N_{\text{eff}}$  matrix;  $n_s = 6$  for  $N_g = 2$ ).

**Step 1: the states.**

$$\phi^I = \frac{x^I}{\|x^I\|}, \quad \rho = X (X^T X)^{-1} X^T, \quad \rho^{IJ} = \rho_{[I N_{\text{eff}}:(I+1)N_{\text{eff}}, J N_{\text{eff}}:(J+1)N_{\text{eff}}]}. \quad (23)$$

**Step 2:  $R^I$  (explicit).**  $R^I$  is a function of  $\phi^I$  alone, Eqs. (3) and (10):

$$\begin{aligned} \Delta_{ab}^{I,\text{bb}} &= \phi^{IT} B_{ab} \phi^I, \quad D_a^I = \frac{1}{4} \phi^{IT} \mathcal{Y}_a \phi^I, \quad \mathbf{I} - \Delta^{I,\text{bb}} = W \text{diag}(d_i) W^T, \\ R^I &= W \left[ (d_i(1 - d_i) + \varepsilon)^{-1/2} \circ (W^T D^I) \right]. \end{aligned} \quad (24)$$

The matrix  $\rho$  plays no role here.  $Z^I = R^{IT} R^I$ .

**Step 3:  $\lambda^I$  (by linear least squares).** The multiplier  $\lambda^I$  is not determined by  $(x, X)$  through a formula but by the two stationarity conditions of Eq. (6), Eqs. (20) and (21), which are *linear* in the entries of  $\lambda^0, \lambda^1$ . Collect them in the unknown vector

$$w = (\lambda_{ab}^0, \lambda_{ab}^1)_{a \leq b} \in \mathbb{R}^{2n_s}. \quad (25)$$

Let  $\kappa_{ab} = 1$  for  $a = b$  and  $\kappa_{ab} = 2$  for  $a < b$  (because  $\sum_{ab} \lambda_{ab} B_{ab} = \sum_{a \leq b} \kappa_{ab} \lambda_{ab} B_{ab}$ ),  $P_I = \mathbf{1} - \phi^I \phi^{IT}$ ,  $K = \begin{bmatrix} 0 & -t R^0 R^{1T} \\ -t R^1 R^{0T} & 0 \end{bmatrix}$  (with the  $R^I$  of Step 2), and let  $\Xi^{(I,ab)}$  be the  $2N_{\text{eff}} \times 2N_{\text{eff}}$  symmetric matrix with 1 at positions  $(I N_{\text{eff}} + a, I N_{\text{eff}} + b)$  and  $(I N_{\text{eff}} + b, I N_{\text{eff}} + a)$  (a single 1 if  $a = b$ ) and zero elsewhere. The linear system is

$$\begin{pmatrix} A^{(0)} & 0 \\ 0 & A^{(1)} \\ C^{(0)} & C^{(1)} \end{pmatrix} w = \begin{pmatrix} b^{(0)} \\ b^{(1)} \\ \text{vec}[K, \rho] \end{pmatrix}, \quad (26)$$

a system of  $2m + 4N_{\text{eff}}^2$  equations for the  $2n_s$  unknowns, with

$$\begin{aligned} A_{\cdot,(ab)}^{(I)} &= 4 \kappa_{ab} P_I (B_{ab} \phi^I) \in \mathbb{R}^m, & b^{(I)} &= P_I g_\phi^{(I)}, \\ C_{\cdot,(ab)}^{(I)} &= \text{vec}[\Xi^{(I,ab)}, \rho] \in \mathbb{R}^{4N_{\text{eff}}^2}, \end{aligned} \quad (27)$$

where  $g_\phi^{(I)}$  is the gradient of  $E$  with respect to  $\phi^I$  at fixed  $\rho$  (Eq. (16), including the dependence through  $R(\phi^I)$ ), and the projector  $P_I$  removes the norm multiplier  $\mu_I$  of Eq. (21). The first two block rows are the  $\phi^I$  conditions, the last block row is  $[\Xi, \rho]$ -form of Eq. (20) ( $[K - \Lambda, \rho] = 0 \Leftrightarrow [\Lambda, \rho] = [K, \rho]$ ). The system is solved in the least-squares sense (`numpy.linalg.lstsq`, `rcond=10-6`, minimum-norm solution), and  $\lambda^I$  is the symmetric matrix built from the  $n_s$  entries of  $w$  belonging to fragment  $I$ .

- *Why jointly.* The  $\phi$  block alone is rank deficient whenever a ghost is inert (then the vectors  $P_I B_{ab} \phi^I$  are linearly dependent), so its minimum-norm solution is not unique and need not satisfy  $[K - \Lambda, \rho] = 0$ ; this happened for the forward solution ( $[H_{\text{qp}}, \rho] \approx 0.16$ ). With the  $C$  block included the system has rank  $2n_s = 12$  for  $N_g = 2$  at the backward and forward solutions examined.
- *Nearly free direction.* For  $U/t \gtrsim 7$  ( $N_g = 2$ ) the smallest singular value of the system is  $\sim 10^{-7}$ , while the next one is  $\sim 0.15$ : one direction of  $\lambda$  is almost undetermined by the stationarity conditions (it belongs to the nearly decoupled bath orbital, whose quasiparticle

level is barely constrained). With a cutoff as small as  $10^{-10}$  that direction is fixed by rounding noise, the residual is unchanged, and the two middle quasiparticle levels jump erratically with  $U$  (values between  $-3$  and  $+4$  were seen). The cutoff `rcond` =  $10^{-6}$  (relative to the largest singular value) removes it, i.e. takes the minimum-norm  $\lambda$ ; results are identical for  $10^{-6}$  and  $10^{-4}$ , and the two levels then vary smoothly and stay at  $\approx 10^{-3}$  (at the Fermi level, as expected for a decoupled bath) with the residual  $\lesssim 8 \times 10^{-7}$ . The position of those two levels is therefore a *choice* of the minimum-norm solution rather than something determined by  $(x, X)$  alone.

- *Quality check.* If  $(x, X)$  is a stationary point, the residual of Eq. (26) vanishes: it is reported as “KKT:  $\phi$  stationarity” (the  $A$  rows) and “KKT:  $[H_{\text{qp}}, \rho]$ ” (the  $C$  rows, recomputed with the final  $H_{\text{qp}}$ ). Values found:  $\lesssim 6 \times 10^{-7}$  and  $\lesssim 6 \times 10^{-8}$  (Table 1). A large residual means that  $(x, X)$  is *not* a stationary point of  $\mathcal{L}$ , and the  $\lambda$  obtained is then only the best least-squares one.

**Step 4: what follows.**  $H_{\text{qp}} = K - \text{diag}(\lambda^0, \lambda^1)$  is built as in `code.tex`, giving the quasiparticle energies `eqp`;  $V^I$  and  $\lambda^{I,c}$  follow from Eqs. 7 and 8 of `code.tex` with  $\rho$ ,  $R^I$ ,  $\Delta^I = \mathbf{I} - \Delta^{I,\text{bb}}$  and  $\lambda^I$ . The arrays  $R^I$ ,  $\lambda^I$ ,  $\lambda^{I,c}$  are written by `save_results` in the layout used by `gga_dimer.py` (two rows of  $N_{\text{eff}}$  numbers for  $R$ ; two stacked  $N_{\text{eff}} \times N_{\text{eff}}$  blocks for each matrix), so the files have the format of `R.in/L.in`.

**Gauge.**  $R^I$  and  $\lambda^I$  are obtained in the basis defined by  $x^I$  (the ghost-orbital basis of the impurity state) and  $X$ ; no gauge fixing is applied. Solutions related by a rotation  $Q_I \in O(N_{\text{eff}})$  of the ghost basis of fragment  $I$  have  $R^I \rightarrow Q_I R^I$  and  $\lambda^I \rightarrow Q_I \lambda^I Q_I^T$ , while  $E$ ,  $Z^I$  and the quasiparticle energies are invariant. (The inverse map, from  $(R, \lambda)$  to a starting  $(x, X)$ , is the seeded start described in Section 5.)

## References

- [1] A. M. Tagliente, I. Pasqua and M. Fabrizio, *Direct minimization versus iterative embedding in the ghost-Gutzwiller method: a comparative study of magnetism in Mott insulators*, arXiv:2607.27156 (`NotImpurity.pdf`); Eqs. 7 and 9–12 are the basis of this note.
- [2] `doc/code.tex`: notation and equations of the embedding solver `gga_dimer.py`, used throughout.